

SURAJ KRISHNA BANAVALIKAR

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Technical Skills

Languages: Java, C/C++, Python, JavaScript

Web Technologies: HTML, CSS, React.js, Node.js, Express.js, REST APIs

Databases: MongoDB, PostgreSQL, MySQL, Pinecone

Developer Tools: Git, GitHub, Docker, Azure cloud, AWS cloud, Linux, Prometheus, Grafana, Postman, Power BI

Education

Ramaiah Institute of Technology, Bangalore

Oct 2023 – Present

B.E. in Computer Science and Engineering

CGPA: 9.19/10

St. Joseph's PU College, Karwar

Aug 2021 – Mar 2023

PCMC (Physics, Chemistry, Mathematics, Computer Science)

Grade: 96.16%

Relevant Coursework

- Data Structures & Algorithms
- Object Oriented Programming
- Database Management Systems
- Artificial Intelligence
- Operating System
- Computer Networks
- Software Engineering
- DevOps and Cloud
- Distributed Systems

Projects

Campus Helpdesk | *Node.js/Express, Python, REST APIs, MongoDB, Pinecone, JavaScript, Docker*

Showcase

- Developed an AI-powered Campus Helpdesk using Retrieval-Augmented Generation (RAG), vector search, and hybrid re-ranking with Pinecone.
- Implemented JWT authentication, role-based authorization, and RESTful APIs for secure access to campus resources.
- Secured the backend with configurable CORS, API rate limiting and scaled with PM2 clustering to support ~150 concurrent users.
- Optimized performance with MongoDB connection pooling and asynchronous job queues (20 jobs per batch with exponential backoff), reducing API blocking and improving throughput under concurrent load.
- Load-tested at 50–60 concurrent users with stable latency, and reduced manual query-handling effort by ~40% based on evaluation against real usage.
- Containerized micro services with Docker, enabling consistent deployments.

StellarRoute | *Python, Leaflet.js, REST APIs, NASA APIs, Redis, Docker, Prometheus, Grafana*

GitHub | Live

- Designed and implemented a GPS-denied navigation system using inertial navigation (IMU-based dead reckoning), processing accelerometer and gyroscope data to estimate position in high-interference environments such as solar storms, without relying on satellite signals.
- Set up a CI/CD pipeline, containerized the application with Docker, and deployed it to AWS as a live application with Prometheus and Grafana for real-time monitoring.
- Integrated NASA's NOVA API for live Kp index values and proposed future integration with ISRO's Aditya-L1 real-time telemetry.

Zero-Server AI Privacy Gateway | *JavaScript, IndexedDB*

GitHub

- Developed a browser extension that detects and sanitizes sensitive information locally before prompts are transmitted to ChatGPT or Claude, ensuring user privacy without sending data to external servers.
- Built a multi-layer PII detection engine using C++ → WebAssembly, regex validation, and heuristic analysis to detect 19+ categories of sensitive information.
- Implemented IndexedDB-based placeholder vault and automatic response restoration using DOM Mutation Observer, ensuring seamless user experience.
- Achieved real-time processing with ~2.86 ms median latency and 92% detection accuracy while maintaining a zero-server architecture.

Experience

RIT S.T.A.R.D.U.S.T Club

May 2025 – Present

Software Engineer — Avionics Member

- Participated in the IROc-U 2026 competition by developing autonomous drone simulations using ROS 2 and Gazebo.
- Assisted in validating simulated drone behavior for competition preparation.

Awards and Certifications

wHACKiest 2025 — 2nd Place

- Secured **2nd Place** among **100+ teams** at the 24-hour intra-college hackathon by developing **StellarRoute** as the Backend Engineer.
- Certificate